You said:

1. What are Tensors
2. 0D Tensor/Scalar
3. 1D Tensor/Vector
4. 2D Tensor/Martices
5. ND Tensors
6. Rank, Axes and Shape
7. Example of 1D Tensors
8. Example of 2D Tensors
9. Example of 3D Tensors
10. Example of 4D Tensors
12. Example of SD Tensors

ChatGPT said:

Tensors in Machine Learning — Easy Explanation

1. What are Tensors?

A **tensor is a container used to store numbers in different dimensions.** In Machine Learning and Deep Learning, tensors store data such as prices, images, videos, and text representations.

2–5. Types of Tensors



0D Tensor — Scalar

One single number: `7`

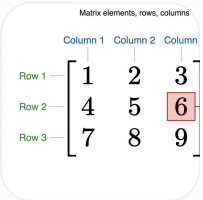
Example: A product's price rating.



1D Tensor — Vector

A list of numbers: `[10, 20, 30]`

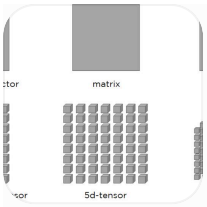
Example: Marks of three students.



2D Tensor — Matrix

Rows and columns: `[[1, 2], [3, 4]]`

Example: Student marks across subjects.



3D and ND Tensors

Data arranged in three or more dimensions.

Example: A color image has height, width, and color channels.

6. Rank, Axes and Shape

Consider this tensor:

```
import numpy as np

x = np.array([[1, 2, 3],
              [4, 5, 6]])

print(x.ndim)  # Rank: 2
print(x.shape) # Shape: (2, 3)
```

- **Rank:** Number of dimensions → `2`
- **Axes:** Directions along dimensions → rows (axis 0), columns (axis 1).
- **Shape:** Size of each dimension → `(2, 3)` means 2 rows and 3 columns.

7–11. Examples of 1D, 2D, 3D, 4D and 5D Tensors

Tensor	Example shape	Real-life use
1D	`(3,)`	3 student marks
2D	`(2, 3)`	2 students × 3 subjects
3D	`(4, 2, 3)`	4 color channels or data groups of matrices
4D	`(10, 32, 32, 3)`	10 RGB images, each 32 × 32 pixels
5D	`(2, 10, 32, 32, 3)`	2 batches of 10 RGB images

12. Example of a 5D Tensor in Python

```
import numpy as np

x = np.zeros((2, 10, 32, 32, 3))

print(x.ndim)   # 5
print(x.shape)  # (2, 10, 32, 32, 3)
```

Remember: 0D = single number, 1D = list, 2D = table, 3D = stack of tables, and higher dimensions = more complex data structures used in Deep Learning.

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12. Example of SD Tensors
explain with very atual values and example easy word
def
concept
code
example

ChatGPT said:

Tensors in Machine Learning — Easy Explanation with Real Values

1. What are Tensors?

Definition: A tensor is a data structure used to store numbers in one or more dimensions.

Concept: Think of tensors as containers for data. A single number, a list, a table, or a collection of images can all be represented as tensors.

Real-life example: In a house-price prediction project, you might store house sizes, bedroom counts, and prices as numerical data.

```
import numpy as np

house_sizes = np.array([500, 1000, 1500])
print(house_sizes)
# [ 500 1000 1500]
```

2. 0D Tensor — Scalar

Definition: A 0D tensor contains just one number.

Concept: It is like one value written on a piece of paper.

Real-life example: A house costs ₹25 lakh.

```
import numpy as np

price = np.array(25)
print(price)        # 25
print(price.ndim)   # 0
print(price.shape)  # ()
```

3. 1D Tensor — Vector

Definition: A 1D tensor stores a sequence of numbers in one dimension.

Concept: Think of a single list of values.

Real-life example: Three houses have sizes of 500, 1000, and 1500 square feet.

```
sizes = np.array([500, 1000, 1500])

print(sizes)
print(sizes.ndim)   # 1
print(sizes.shape)  # (3,)
```

4. 2D Tensor — Matrix

Definition: A 2D tensor contains values arranged in rows and columns.

Concept: Think of an Excel table.

Real-life example: Three houses, each with size and number of bedrooms.

```
houses = np.array([
    [500, 1],
    [1000, 2],
    [1500, 3]
])
```

```
print(houses)
print(houses.ndim)   # 2
print(houses.shape)  # (3, 2)
```

Here, each row represents a house, and the two columns represent size and bedrooms.

5. ND Tensors — N-Dimensional Tensors

Definition: ND means a tensor can have any number of dimensions.

Concept: As dimensions increase, data can be organized into more complex structures.

Real-life example: A video contains multiple frames, and each frame contains image pixels.

```
video = np.zeros((10, 3, 64, 64))

print(video.ndim)   # 4
print(video.shape)  # (10, 3, 64, 64)
```

This example represents 10 RGB images, each with 64 × 64 pixels. The shape order here is batch, channels, height, width.

6. Rank, Axes and Shape

NumPy provides `ndim`, `shape`, and `size` to inspect an array. NumPy v2.5 ... +1

```
x = np.array([[10, 20, 30],
              [40, 50, 60]])

print(x.ndim)   # 2: number of dimensions
print(x.shape)  # (2, 3): 2 rows, 3 columns
print(x.size)   # 6: total numbers
```

- **Rank:** Number of dimensions; here it is 2.
- **Axes:** Directions in the array. Axis 0 refers to rows; axis 1 refers to columns.
- **Shape:** The size of each dimension, here `(2, 3)`.

7. Example of a 1D Tensor

```
marks = np.array([75, 80, 95])
```

```
print(marks[0])  # 75
print(marks.shape)  # (3,)
```

Use: Store marks of three students.

8. Example of a 2D Tensor

```
marks = np.array([[75, 80],
                  [85, 90],
                  [95, 88]])

print(marks[1, 0])  # 85
print(marks.shape)  # (3, 2)
```

Use: Three students, two subject marks each.

9. Example of a 3D Tensor

```
data = np.array([
    [[10, 20], [30, 40]],
    [[50, 60], [70, 80]]
])

print(data.shape)  # (2, 2, 2)
print(data[0, 1, 0])  # 30
```

Use: Two groups, each containing a 2 × 2 table.

10. Example of a 4D Tensor

```
images = np.zeros((5, 3, 32, 32))

print(images.ndim)   # 4
print(images.shape)  # (5, 3, 32, 32)
```

Use: A batch of 5 RGB images, each 32 × 32 pixels.

11. Example of a 5D Tensor

Your list says “SD”; assuming you mean **5D tensor**:

```
videos = np.zeros((2, 10, 3, 32, 32))

print(videos.ndim)   # 5
print(videos.shape)  # (2, 10, 3, 32, 32)
```

Use: 2 video clips, 10 frames per clip, 3 color channels, and 32 × 32 pixels per frame.

Quick revision

Dimension	Easy meaning	Example
0D	Single number	`25`
1D	List	`[10, 20, 30]`
2D	Table	`[[1, 2], [3, 4]]`
3D	Stack of tables	Groups of matrices
4D	Batch of images	Images with color channels
5D	Batch of videos	Clips containing frames

Key point: `ndim` tells you how many dimensions exist, while `shape` tells you the size of each dimension.